

Constraining spatial variability of methane ebullition seeps in thermokarst lakes using point process models

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Received 28 August 2012; revised 14 June 2013; accepted 16 June 2013; published 19 July 2013.

[1] Ebullition is an important but highly heterogeneous mode of methane emission in lakes. Variability in both spatial distribution and temporal flux creates difficulty in constraining uncertainties in whole lake emission estimates. Analysis of short- and long-term flux measurements on 162 ebullition seeps in 24 panarctic lakes confirmed that seep classes, identified a priori according to bubble patterns in winter lake ice, have distinct associated fluxes irrespective of lake or region. To understand the drivers of ebullition's spatial variability and uncover ways to better quantify ebullition in field work, we combined point-process modeling with field measurements of 2679 GPS-marked and classified ebullition seeps in three Alaskan thermokarst (thaw) lakes that varied by region, permafrost type, and seep distribution. Spatial analysis of field data revealed that seeps cluster above thawed permafrost soil mounds in lake bottoms. Seep density and clustering, determined from field observations, were used as parameters in a Poisson cluster process model to simulate seeps across entire lake surfaces. Sampling results indicated that (1) applying seep-class mean flux values to unmeasured seeps counted on ice-bubble surveys does not compromise accuracy of whole lake flux estimates; (2) three distributed 50 m² ice-bubble survey transects more accurately estimate mean lake ebullition than 17 dispersed 0.2 m² bubble traps; and (3) the uncertainty associated with whole lake mean ebullition estimated by lake-ice survey transects is inversely related to seep density. Findings suggest that transect field data collected on a large number of widely distributed lakes can be combined to provide a well-constrained, bottom-up estimate of regional lake ebullition.

Citation: Walter Anthony, K. M., and P. Anthony (2013), Constraining spatial variability of methane ebullition seeps in thermokarst lakes using point process models, *J. Geophys. Res. Biogeosci.*, 118, 1015–1034, doi:10.1002/jgrg.20087.

1. Introduction

[2] Methane cycling in aquatic ecosystems involves microbial methanogenesis in anaerobic sediments, diffusion of aqueous methane through sediments and water, and aerobic and anaerobic methane oxidation [Valentine and Reeburgh, 2000; Valentine et al., 2001; Liikanen et al., 2002]. Other important components of methane cycling and transport include bubble formation and migration in sediments, bubble release at the sediment-water interface, partial or full dissolution of bubbles during their rise through the water column, and potential bubble release at the water surface to the atmosphere [Martens and Klump, 1980; Chanton et al., 1989; Joyce and Jewell, 2003; Boudreau et al., 2001, 2005; McGinnis et al., 2006; Scandella et al., 2011a, 2011b].

[3] In many lakes, ebullition (bubbling) is the dominant mode of methane emissions [Crill et al., 1988; Keller and Stallard, 1994; Casper et al., 2000; Bastviken et al., 2004, 2011; Walter et al., 2006; Del Sontro et al., 2010, 2011; Schubert et al., 2012]. However, spatial and temporal heterogeneity in the ebullition process gives rise to large uncertainties in estimating ecosystem methane emissions [Del Sontro et al., 2010; Wik et al., 2011a]. For instance, ebullition is episodic and not representatively captured by typical short-term measurements [Bastviken et al., 2004; Varadharajan, 2009], a problem that is frequently encountered in sampling of rare ecological events [Thompson, 2012].

[4] In many northern lakes, the spatial clustering and temporal variability in the flux of discrete bubbling point sources, herein called “ebullition seeps,” contribute to heterogeneity in lake ebullition. While ebullition seeps release bubbles episodically, with frequencies of several minutes to weeks depending on the seep type, atmospheric pressure dynamic, and season of year, the location of many seeps in lakes with dense sediments remains stable over time scales of at least seasons to years. Walter Anthony et al. [2010] marked the location of 17 discrete ebullition seeps in Siberian and Alaskan lakes and found that the majority of seeps did not change in location over observation periods of 2 and 8 years. Temporal consistency in the location of ebullition seeps is likely the result of formation of fractures

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2169-8953/13/10.1002/jgrg.20087

	Seep class	Description	Ebullition (ml gas d ⁻¹)	
			Mean $\pm$ s.e.m., <i>n</i> seeps	Min-max
				
	A	Stacks of small individual bubbles, >50% unmerged; usually <40 cm diameter	22 $\pm$ 4, <i>n</i> =30	2-96 (0-47)
	B	Bubbles >50% merged, clustered in multiple layers of ice; usually >25 cm diameter	211 $\pm$ 39, <i>n</i> =42	32-1,081 (0-1,030)
	C	Single large pockets of near 100% merged bubbles stacked in ice; occasional, isolated bubbles around <25% of perimeter; usually >40 cm diameter	1,726 $\pm$ 685, <i>n</i> =35	525-2,649 (0-24,640)
	Hotspot	Ice-free hole in lake ice due to frequent bubbling. Thin snow or ice crust, easily broken with fist or foot, often covers the hole.	7,801 $\pm$ 764, <i>n</i> =55	3,110-16,918 (1,313-25,412)

Figure 1. Ebullition seep classes (A, B, C, and Hotspot) distinguished according to visible patterns of bubbles trapped in lake ice and the associated bubbling rates that generate these patterns. Error term represents the standard error of the mean (s.e.m.). The minimum and maximum daily ebullition rates are shown for long-term measured seeps (no parentheses) and for short-term measured seeps (inside parentheses). The top left photograph shows a 1 m wide bubble survey transect on lake ice cleared of snow. The diameters of the example A, B, C-type bubble clusters in the photographs, as shown by the white lines, are 20 cm, 25 cm, and 45 cm respectively. The open-hole Hotspot has a 30 cm diameter. This classification scheme, presented previously by *Walter et al.* [2006] and *Walter Anthony et al.* [2010], is shown here with additional new flux data from 20 lakes and greater detail in the distinguishing seep characteristics.

or “bubble tubes” in the fine-grain sediments, which have been described previously in other field studies [*Martens et al.*, 1980.; *Martens and Klump*, 1980; *Boudreau et al.*, 2005; *Scandella et al.*, 2011a] and lab experiments [*Scandella et al.*, 2011b]. When a sufficient volume of gas is produced or when the hydrostatic pressure drops enough to dislocate large bubbles in sediments, the bubbles break out creating preferential flow channels in the sediments [*Varadharajan*, 2009]. Once these fractures are formed, escape of bubbles continues through the bubble tubes, resulting in consistency in the location of bubble seepage from the lake bottom.

[5] During ice-free conditions, methane-rich bubbles released from lake sediments constitute direct emissions to the atmosphere in lakes where water depth and bubble size allow for bubbles to survive their ascent to the lake surface [*McGinnis et al.*, 2006], which was the case for all of the >100 panarctic lakes where ebullition bubbles have been recently investigated [*Wik et al.* 2011a, 2011b; *Brosius et al.*, 2012; *Walter Anthony et al.*, 2010, 2012] and 20 lakes

in East Antarctica [*Sasaki et al.*, 2009]. When lake ice forms, rising bubbles are trapped by the bottom of the lake-ice sheet. Throughout most of the winter, methane-rich bubbles freeze into the ice, becoming sealed in the ice when the downward growing lake ice thickens around them [*Gow and Langston*, 1977]. *Walter et al.* [2006] used distinctions in the patterns that seep bubbles form in winter lake ice to classify discrete ebullition seeps into four types: (A) kotenok, (B) koshka, (C) kotara, and Hotspots. These relatively large ebullition bubbles (1cm to >100 cm), rich in methane (up to 95%) are distinct from tiny (millimeter scale), tubular ice-bubbles with lower methane concentrations (usually <1%) that result from the outfreezing of dissolved gases [*Boereboom et al.*, 2012]. A-type ebullition seeps are relatively small clusters of bubbles in which individual bubbles stack on top of each other in ice without merging laterally (Figure 1). Due to relatively higher ebullition rates, individual bubbles released from sediments in B-type seeps laterally merge into larger bubbles under the ice prior to freezing in ice. Types A and B seeps produce low gas-volume clusters of bubbles in lake

ice (Figure 1), whereas larger C seeps result in big pockets of gas in ice separated vertically by ice layers containing few or no bubbles. The solid ice layers in between the large gas pockets of C-type seeps represent periods of relative quiescence in between large ebullition events. The frequency of ebullition release from Hotspot seeps and the associated convection in the water column created by rising bubble plumes can be strong enough to maintain mostly ice-free holes in winter lake ice (Figure 1). Not all lakes have ebullition Hotspots. In thermokarst (thaw) lakes, Hotspot seeps are typically concentrated along margins of most active permafrost thaw [Walter *et al.*, 2006], where labile organic matter is made abundantly available to methanogenesis by deep permafrost thaw in taliks [Kessler *et al.*, 2012]. The mean annual methane ebullition effluxes from sediments associated with each of the seep types were recently reported based on year-round, long-term continuous flux measurements of ebullition seeps in four Siberian and Alaskan thermokarst lakes [Walter Anthony *et al.*, 2010].

[6] While some progress has been made toward constraining heterogeneity in ebullition seep classification and temporal effluxes from sediments, the pattern of spatial variability in ebullition is unresolved. Taking advantage of trapped bubbles in lake ice, researchers follow a relatively new method outlined by the Pan-Arctic Lake Ice Methane Monitoring Network (PALIMMN, <http://ine.uaf.edu/werc/palimmn/>) and described in detail by Walter Anthony *et al.* [2010] to examine the spatial distribution of bubbling events in lakes. In ice-bubble surveys, transects are placed randomly in different zones of the lake (stratified randomization) to account for different shore-type processes, such as thermokarst erosion, and other sources of geographic variability that could influence ebullition within lakes. How representative these transect-derived field survey data are of the true whole-lake ebullition flux from sediments is unknown because a very limited fraction of the lake is surveyed. Researchers who used this method found that the spatial variability of seeps within individual lakes was as high as 10^2 – $10^3\%$ in Alaskan, Siberian, and Swedish lakes [Walter Anthony *et al.*, 2010; Wik *et al.*, 2011a]. Confounding this problem, stratified randomization in the placement of bubble traps in summer resulted in 4–5 times lower emission estimates than those produced by ice-bubble surveys [Walter *et al.*, 2006; Zimov *et al.*, 1997; Wik *et al.* 2011a, 2011b]. Thus, the accuracy of each method for estimating flux and the causes of discrepancy between methods need to be resolved.

[7] Due to the spatial and temporal heterogeneity in ebullition efflux from sediments and the tendency for total ebullition flux to be dominated by events that are rare in space and time, accurate estimation of ebullition faces problems similar to those faced when sampling other rare ecological events. These problems include the near-impossibility of systematic ground sampling to observe every ebullition seep and the tendency of classical random sampling methods (e.g., stratified randomization in the placement of bubble traps) to produce sampling distributions that underestimate the true mean [Thompson, 2012].

[8] Since we cannot know the entire real population of ebullition seeps on a lake, field estimates of ebullition flux cannot be compared to a known mean lake flux. In this situation, stochastic simulation provides the best method for

evaluating the effectiveness of various field methods for estimating ebullition [Thompson, 2012]. Spatial point process models allow characterization of the spatial patterns of ebullition. In ice-bubble ebullition surveys, seep locations are identified in lakes using global positioning systems (GPS) and classified as A, B, C, or Hotspot based on ice bubble morphology (Figure 1). Each seep has a location and a classification, and so may be considered as a marked point process [Cressie, 1993; Bivand *et al.*, 2008]. Coupling point process models of seep location and class with field measurements of seep flux allows simulation of realistic populations of ebullition seeps and simulation-based evaluation of ebullition sampling strategies.

[9] The objectives of this study were to use spatially explicit, individual seep field data combined with point-process models to describe and interpret spatial patterns of ebullition in three intensively studied thermokarst lakes located in different permafrost types and regions of Alaska, and to evaluate field methods for quantifying ebullition. First we test the hypothesis that specific seep classes (A, B, C, and Hotspot) identified according to ice-bubble patterns differ by ebullition rate irrespective of lake type or region. Then, we investigate the nature and causes of spatial heterogeneity in methane ebullition in thermokarst lakes, particularly as they relate to permafrost thaw geomorphology. We combine our understanding of the spatial pattern of ebullition with other field observations of bathymetry and extent of thermokarst erosion along lake margins to simulate ebullition across the whole lake surfaces with point process modeling. Using iterated and randomized sampling of these simulated ebullition data sets, we compare the accuracies of the ice-bubble survey transects and the classical random sampling by bubble traps methods for estimating whole-lake ebullition release from sediments. By quantifying the errors associated with transect-based lake ebullition surveys, we provide a solution for reducing uncertainty in regional-scale lake ebullition estimates based on limited field data.

2. Methods

2.1. Site Description

[10] This study focused on three thermokarst lakes located in different regions of Alaska (Figure 2) that were unglaciated during the late Pleistocene [Kaufman and Hopkins, 1986; Manley and Kaufman, 2002]. The lakes differed in size, depth, and their topographic position on the landscape. Lake depths varied according to the distribution of excess ground ice in permafrost in which they formed.

[11] Lake Claudi (informal name; 66.55°N, 164.45°W; 162,750 m²), located in the continuous permafrost zone on the northern Seward Peninsula, is a 10 m deep lake formed in yedoma-type permafrost. Yedoma was recognized first as an extensive permafrost type in Russia [Birkengof, 1933; Tomirdiario, 1980; Kaplina, 1981], characterized by ice-supersaturated Pleistocene-aged loess and loess-like deposits. Thick yedoma deposits in Siberia were reported to store large quantities of organic carbon (~450 Gt) [Zimov *et al.*, 2006], which can contribute to production and emission of methane and carbon dioxide upon permafrost thaw [Zimov *et al.*, 1997]. Yedoma-type permafrost also occurs extensively in Alaska [Kanevskiy *et al.*, 2011] and has been described on the northern Seward Peninsula as late-Pleistocene-aged

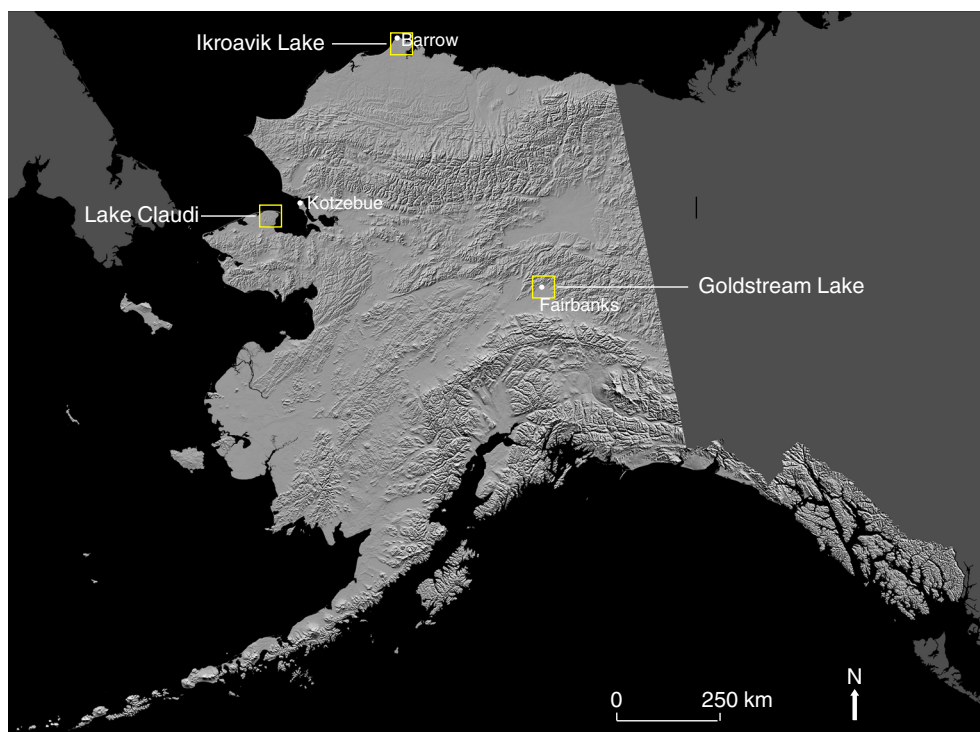


Figure 2. Map of Alaska showing the location of study lakes (yellow squares) on the northern Seward Peninsula (Lake Claudi), interior Alaska (Goldstream Lake), and the coastal plain of the Alaska North Slope (Ikroavik Lake). The locations of nearby communities at Barrow (71.29°N, 156.79°W), Kotzebue (66.90°N, 162.60°W), and Fairbanks (64.88°N, 147.82°W) are shown as white dots. The Alaska map is the National Elevation Data Set 30 m hillshade raster.

aeolian silt with deep syngenetic ice in the form of ice-wedges, lenses and pore ice that comprise 60–80% of the substrate by volume [Hopkins *et al.*, 1955, Hopkins and Kidd, 1988; Hoeffle *et al.*, 2000]. Permafrost depth on the northern Seward Peninsula generally exceeds 90 m [Hopkins *et al.*, 1955]. Thermokarst expansion was observed along all margins of Lake Claudi during recent decades; however, the most rapid expansion occurred along the NE and SW shores [Jones *et al.*, 2009, 2011], as shown in Figure 3a. The bathymetry, morphology, and numerical modeling of Lake Claudi's expansion, parameterized by site-specific field data, were presented recently by West and Plug [2008] and Kessler *et al.* [2012].

[12] Goldstream Lake (informal name; 64.92°N, 147.85°W; 10,030 m²; 2.9 m deep), located in the discontinuous permafrost zone of interior Alaska near Fairbanks, formed in retransported late-Quaternary loess that is common on many hill slopes and valley bottoms of interior Alaska [Muhs and Budahn, 2006]. Unlike the majority of the yedoma lowlands in Siberia [Zimov *et al.*, 2006] and the northern Seward Peninsula [Hopkins *et al.*, 1955, Hopkins and Kidd, 1988], where permafrost formed syngenetically (concurrent with deposition) during the late Pleistocene, colluvial forces and frost action in interior Alaska gradually eroded loess down slope during the late Pleistocene and early Holocene, forming icy, organic-rich deposits described as “muck,” frequently several tens of meters to 100 m thick in valley bottoms [Péwé, 1975; Hamilton *et al.*, 1988; Muhs and Budahn, 2006; Reyes *et al.*, 2010]. Massive ice wedges, 2 to 4 m wide at their tops are common [Hamilton *et al.*,

1988]. Based on their high ice content, large syngenetic ice wedges, cryostructure, presence of mammoth fauna remains, often high organic carbon contents (0.38–6.8% C) [Hamilton *et al.*, 1988], and occurrence of distinct thermokarst landforms, Kanevskiy *et al.* [2011] classified these interior Alaska deposits as “yedoma-type.” Remote-sensing based observation of lake extent shows that a partial drainage event occurred in Goldstream Lake sometime after 1949 (Figure 3b); however, we observed that thermokarst expansion continues presently, predominately along the eastern margin of Goldstream Lake.

[13] Ikroavik Lake (71.23°N, 156.63°W; 5,292,850 m²), the largest of the study lakes, has a maximum depth of 2.4 m and is located on the Barrow Peninsula on the Arctic Coastal Plain of northern Alaska. The region is underlain by continuous permafrost with maximum thickness in excess of 400 m [Hinkel *et al.*, 2003]. Like most of the lakes on the Barrow Peninsula, Ikroavik Lake is elliptical (Figure 3c), with the major axis oriented a few degrees west of due north and nearly perpendicular to the prevailing summer wind direction [Carson and Hussey, 1962; Sellmann *et al.*, 1975]. Barrow Peninsula lakes developed in ice-rich permafrost [Livingstone *et al.*, 1958; Black, 1969]; however, their shallow depth is in part the consequence of numerous iterations of the thermokarst-lake cycle in the region [Hinkel *et al.*, 2003, 2007; Eisner *et al.*, 2005] that limits the volume and distribution of excess ground ice in which thermokarst lakes form today [Jorgenson and Shur, 2007]. Very few remnants of thicker ice-rich units remain [Eisner *et al.*, 2005; Frohn *et al.*, 2005]. The shores types of

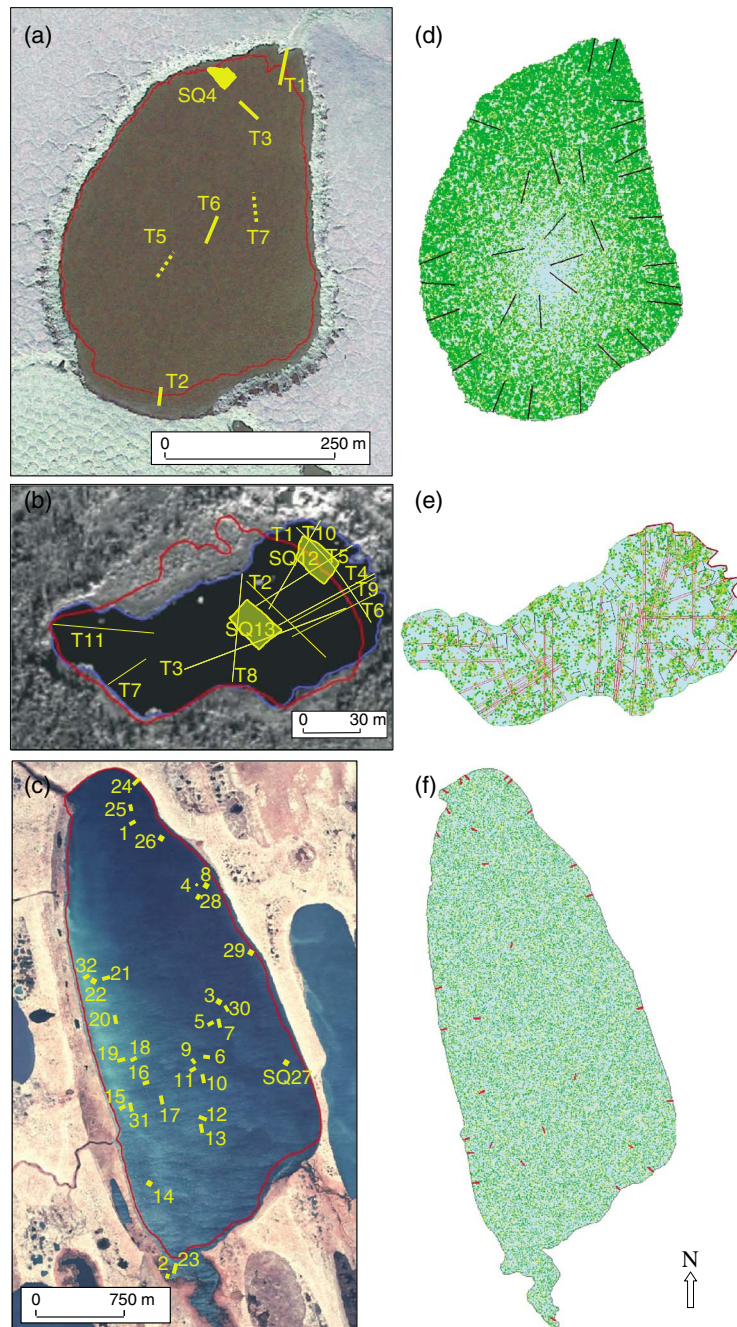


Figure 3. The locations of 2007–2011 field survey transects and simulated ebullition seeps on Lake Claudi, Goldstream Lake, and Ikroavik Lake. Field survey transects are drawn to scale overlying recent satellite images on (a) Lake Claudi, 2008 pan-sharpened, multispectral IKONOS© image with a resolution of 1 m, and on (b) Goldstream Lake, 27 August 2008 Worldview image with a resolution of 0.5 m. The approximate locations of transects for which precise GPS positions were unavailable are shown as dashed lines (Claudi T5, T7). Field survey transects are not to scale on (c) Ikroavik Lake, CIR photo, 2.5 m resolution; actual survey areas are provided in Table 2. The previous shoreline positions, shown as red lines, were from 1955, 1949, and 1951 black and white aerial photographs with ≈ 1 m resolution for Lake Claudi, Goldstream Lake, and Ikroavik Lake, respectively, based on Jones *et al.* [2009]. Baydjarakhs (conical thermokarst mounds), now covered in shrub vegetation are seen across the bottom of a drained thermokarst lake basin adjacent to Lake Claudi in the bottom left corner of Figure 3a. Baydjarakh spacing of approximately 10 m demonstrates the morphology of the Pleistocene yedoma ice-wedge network. Baydjarakhs also cover the bottom of Lake Claudi (Figure 7). (d–f) Lake Claudi_{sim}, Goldstream_{sim} Lake, and Ikroavik_{sim} Lake simulated by point process modeling using the ratios, distributions, and clustering of ebullition seep types A (green dots), B (yellow dots), C (orange dots), and Hotspot (red dots) from field observations are also shown. The randomly placed transects and polygons on simulated lakes are drawn to scale.

Table 1. Seep Density and Estimated Ebullition Flux from GPS Field Surveys During 2007–2011 on Lake Claudi, Goldstream Lake and Ikroavik Lake

									Ebullition on Transects	Whole- Lake Ebullition
Field Date	Transect	Area (m ²)	Seep Density (seeps m ⁻²)					Percent of Lake Surveyed	(mL gas m ⁻² d ⁻¹)	
			A	B	C	HS	All Seeps			
<i>Lake Claudi</i>										
	Whole lake		1.43	0.28	0.04	0	1.74	0.57%		152 ± 38
3 Nov 2008	T1	62	2.72	0.47	0.10	0	3.29		326	
3 Nov 2008	T2	33	0.88	0.40	0	0	1.28		103	
6 Nov 2008	T3	37	1.02	0.43	0	0	1.45		112	
4 Nov 2008	SQ4	639	1.68	0.31	0.04	0	2.03		175	
19 Apr 2009	T5	50	0	0	0	0	0		0	
21 Apr 2009	T6	50	0.02	0.04	0	0	0.06		9	
21 Apr 2009	T7	50	0.02	0	0	0	0.02		0	
<i>Goldstream Lake</i>										
	Thermokarst zone		0.88	0.42	0.14	0.09	1.53	10.74%	1,064	
	Nonthermokarst zone		0.25	0.04	0.02	0.00	0.31		75	
	Whole lake, zone-weighted		0.36	0.11	0.04	0.02	0.53		0	253 ± 82
	Whole lake, transects only		0.54	0.18	0.04	0.02	0.78		0	254 ± 62
12 Oct 2007	T1	71	0.40	0.43	0.06	0.07	0.95		750	
12 Oct 2007	T2	87	0.01	0.05	0	0.01	0.07		99	
12 Oct 2007	T3	61	0.13	0.05	0.02	0.02	0.21		169	
20 Oct 2009	T4	50	0.94	0.24	0.08	0	1.25		208	
20 Oct 2009	T5	58	0.70	0.17	0.03	0.02	0.92		244	
20 Oct 2009	T6	68	1.36	0.22	0.01	0	1.60		102	
26 Oct 2009	T7	30	0.43	0.10	0.03	0	0.57		88	
26 Oct 2009	T8	50	0.67	0.14	0.04	0	0.85		112	
19 Oct 2010	T9	48	0.84	0.23	0	0.04	1.11		393	
19 Oct 2010	T10	38	0.72	0.37	0.11	0.03	1.22		485	
20 Oct 2010	T11	54	0.04	0.06	0.07	0	0.17		140	
29 Oct 2011	SQ12	225	1.03	0.41	0.17	0.10	1.71		1,162	
30 Oct 2012	SQ13	236	0.30	0.01	0.00	0	0.31		16	
<i>Ikroavik Lake</i>										
	Whole lake		0.07	0.01	0	0	0.07	0.03%	0	2.8 ± 1.3
28–29 Oct 2009	T1 and T2	37	0	0	0	0	0		0	
27 Oct 2011	T3 - T22	703	0	0	0	0	0		0	
29 Oct 2009	T23	72	0.01	0	0	0	0.01		0	
28 Oct 2009	T24	34	0.09	0.06	0	0	0.15		14	
28 Oct 2009	T25	33	0.12	0.12	0	0	0.24		28	
25 Oct 2011	T26	136	0.01	0	0	0	0.01		0	
27 Oct 2011	SQ27	276	0.03	0	0	0	0.03		1	
26 Oct 2011	T28	161	0.03	0	0	0	0.03		1	
26 Oct 2011	T29	361	0.27	0.02	0	0	0.29		9	
27 Oct 2011	T30	26	0.04	0	0	0	0.04		1	
27 Oct 2011	T31	19	0.05	0	0	0	0.05		1	
27 Oct 2011	T32	17	0.06	0	0	0	0.06		1	

Ikroavik Lake were described by Wohlschlag [1953] as consisting of gravel that grades into sand at the northeastern beach, a 1.5 m tundra bluff along the western margin that is formed by ice movement in spring, and swampy shoreline around the remainder of the lake. Expansion of the lake, caused by wind and wave action and thermal erosion occurred predominately along the eastern and southeastern margins (Figure 3c).

2.2. Field Sampling and Analysis

[14] The three lakes described above were surveyed intensively for spatial bubble patterns. In addition to transects, we surveyed large polygons, referred to as “squares” in these lakes, which allowed quantification of statistical parameters for spatial modeling the lakes. Extensive ice-bubble surveys were conducted on lakes in the northern Brooks Range region, near Toolik Field Station, Alaska. Bubble-trap flux measurements on 162 seeps from 24

panarctic lakes were used to scale the bubble surveys to estimates of flux. We also collected and radiocarbon dated bubble gas to help infer its source.

2.2.1. Ice-Bubble Surveys to Estimate Whole-Lake Seep Ebullition

[15] We followed a stratified randomization sampling design for surveying ebullition seeps in winter lake ice at Lake Claudi, Goldstream Lake, and Ikroavik Lake. In November 2008, we cleared snow from lake ice in a 639 m² polygonal plot, called “Claudi Square,” offshore at the northern margin of Lake Claudi. Each of the 1076 seeps in the plot was categorized as A, B, C, or Hotspot using the seep classification method shown in Figure 1 and was mapped using a centimeter-accuracy RTK differential GPS (Leica Geosystems AG). As covariate data, we mapped the bathymetry of the lake bed with an intensity of ≈0.75 measurements per m² using sonar (Vexilar LPS-1 Hand-Held Depth Finder) point measurements through ice. Approximately square

Table 2. Geographic Information for 24 Lakes on Which Ebullition Flux Measurements Were Conducted^a

Region	Lake	Permafrost	Ecology	Latitude	Longitude	Depth (m)	Number of Flux-Measured Seeps
Alaska, Arctic Coastal Plain	Cake Eater*	continuous	tundra	71.279	-156.637	1.8	8
Alaska, Arctic Coastal Plain	Ikroavik*	continuous	tundra	71.247	-156.641	2.4	5
Alaska, Seward Peninsula	Rhonda	continuous	tundra	66.566	-164.466	1.5	12
Alaska, Seward Peninsula	Cocker Gap	continuous	tundra	66.562	-164.451	5.4	4
Alaska, Seward Peninsula	Fox Den	continuous	tundra	66.559	-164.456	2.4	2
Alaska, Seward Peninsula	Claudi	continuous	tundra	66.552	-164.450	11.0	7
Alaska, Seward Peninsula	Kim	continuous	tundra	66.516	-164.251	3.3	10
Alaska, Seward Peninsula	Jaeger	continuous	tundra	66.501	-164.260	13.3	19
Alaska, Seward Peninsula	Three Loon	continuous	tundra	66.497	-164.254	9.0	5
Alaska, Interior	Vault	discontinuous	boreal forest	65.029	-147.699	5.0	2
Alaska, Interior	Goldstream	discontinuous	boreal forest	64.916	-147.848	2.9	28
Alaska, Interior	Doughnut	discontinuous	boreal forest	64.899	-147.910	2.5	3
Alaska, Interior	Killarney	discontinuous	boreal forest	64.870	-147.902	2.0	8
Alaska, Interior	Smith*	discontinuous	boreal forest	64.865	-147.866	4.0	1
Alaska, Interior	Deuce*	discontinuous	boreal forest	64.864	-147.940	6.0	3
Alaska, Interior	Stevens Pond	discontinuous	boreal forest	64.863	-147.871	>0.8	2
Alaska, Interior	Ace*	discontinuous	boreal forest	64.862	-147.936	9.0	8
Alaska, Interior	Rosie Creek beaver pond	discontinuous	boreal forest	64.770	-148.079	3.9	2
Alaska, Southcentral	Tern	discontinuous	tundra	63.398	-148.670	12.5	4
Canada, Yukon Territories	Dredge Pond	discontinuous	boreal forest	64.035	-139.281	2.0	1
West Greenland	G11	continuous	tundra	67.056	-50.441	>1	1
Russia, Kolyma Lowlands	Grass	continuous	boreal forest	68.752	161.377	12.0	5
Russia, Kolyma Lowlands	Shuchi*	continuous	boreal forest	68.746	161.393	11.0	20
Russia, Kolyma Lowlands	Tube Dispenser*	continuous	boreal forest	68.764	161.403	17.0	2

^aLake names are informal, except those formal names indicated by asterisk.

polygons were selected so that seep observations would be present across a range of x and y axes values; polygon boundaries were also identified with RTK GPS for use in spatial analysis. Ebullition seeps within six additional 1 m wide survey transects that varied in length from 33 to 62 m (Table 1) were mapped in Lake Claudi in either November 2008 or April 2009, the lake ice sheet of a single winter season. Since lake ice freezes from the top down, surveying ebullition seep bubbles in the top 30 to 40 cm of lake ice in April reveals the ebullition patterns that formed in the fall of the year prior. Therefore, the November 2008 and April 2009 data sets can be directly compared as spatial data sets of contemporary ebullition activity. Transects in Lake Claudi were placed so as to originate from different shore types (virgin yedoma, T1; nonvirgin yedoma, T2), off-shore (T3, SQ4), and in the central portions of the basin (T5-7) (Figure 3a).

[16] In Goldstream Lake, we mapped the location of ebullition seeps along eleven 1 m wide transects at a frequency of three to five transects per year in October 2007, October 2009, and October 2010 (Figure 3b) using Garmin 76CSx GPS with Wide Area Augmentation System differential correction. Transects varied by length and seep density (Table 1). Given minimal spatial overlap among transects and observations of interannual seep-position stability, spatial information from all transects in all years was pooled into a single data set. In October 2011 we mapped 303 ebullition seeps within two large polygons using a centimeter-accuracy RTK differential GPS (Leica Geosystems AG). Goldstream Thermokarst Square (SQ12, 225 m²) was located an average of 7 m off shore along the eastern thermokarst shore of the lake (Figure 3b). Goldstream Center Square (236 m²) was located outside the thermokarst zone in the approximate center of Goldstream Lake, where the density of methane ebullition seeps was

significantly lower. As covariate data, we mapped the bathymetry of the lake bed with a 0.22 m² point measurement density on both Goldstream squares.

[17] We classified and mapped ebullition seeps with a Garmin 76CSx GPS along five transects in Ikroavik Lake on 28–29 October 2009. In October 2011 we surveyed ebullition seeps and mapped bathymetry in 27 additional survey plots, including Ikroavik Square, a large 276 m² polygon. Survey plots were randomly located across the lake, but constrained to areas where wet snow had not frozen to the lake-ice surface. We used centimeter-accuracy RTK differential GPS (Leica Geosystems AG) for 2011 surveys. Locations of the 32 field survey plots on Ikroavik Lake are shown in Figure 3c.

[18] To relate field observations of ebullition-seep clustering in lake ice to patterned permafrost ground, we mapped the distribution of ice wedge polygons and baydjarakhs [conical thermokarst mounds] along the eastern margin of Goldstream Lake and along the margins of Lake Claudi in summer. Field work at Ikroavik Lake took place only in winter, at a time when snow cover inhibited accurate mapping of ice wedge polygons in the surrounding tundra.

[19] Finally, we surveyed ebullition along 64 transects in 13 other lakes in the northern foothills of the Brooks Range near Toolik Field Station (68.4°N, 149.4°W). The number of randomly placed, dispersed transects per lake ranged from two to nine, with median and mean values of four and five transects per lake, respectively. These transects will be used as an example of how to apply the results from the spatial surveys and error estimation from the modeling.

2.2.2. Seep Gas Collection and Radiocarbon Dating of Methane

[20] We collected gas samples of ebullition bubbles from 49 seeps at the lake surface in Goldstream Lake during 2008–2011 and Lake Claudi during a 2 week expedition in

October 2008 using submerged bubble traps. We followed methods described previously for gas collection and laboratory analyses [Walter Anthony *et al.*, 2012; Stuiver and Polach, 1977], including bubble methane concentrations and radiocarbon dating methane collected from two ice-bubble pockets above baydjarakhs on Claudi Square.

2.2.3. Ebullition Flux Measurements Using Submerged Bubble Traps

[21] We measured ebullition flux on short-term or long-term time scales on 162 individual seeps classified a priori as A, B, C, or Hotspot based on ice-bubble patterns (Figure 1) on 24 lakes located in Siberia, Alaska, northwest Canada, and west Greenland (Table 2). Twenty-five percent of the 162 panarctic seeps were located in our intensive study lakes, Lake Claudi (seven seeps), Goldstream Lake (28 seeps), and Ikroavik Lake (five seeps). This effort to quantify seep ebullition in panarctic lakes is a component of the Pan Arctic Lake Ice Methane Monitoring Network (PALIMMN). Flux data from four of the 24 lakes reported on here were included in previous publications: Three lakes in Siberia and one lake in Alaska [Walter *et al.*, 2006; Walter Anthony *et al.*, 2010]. This study reports new long-term and short-term ebullition flux data from 20 additional lakes in Alaska, Canada, and Greenland. Altogether this data set consists of $\approx 213,600$ individual ebullition flux measurements made using submerged bubble traps on 162 seeps in 24 panarctic lakes. These data represent rates of bubble release from sediments, measured at the lake surface during multiple seasons of the year. Since winter ice impedes the majority of ebullition bubbles from reaching the atmosphere during the ice cover season, the rates presented here do not represent seasonal ebullition fluxes to the atmosphere. The latter requires consideration of gas exchange between lake water and bubbles trapped under ice and lake-ice phenology.

[22] Prior to measuring flux, seeps were identified as bubbles in ice and classified based on bubble morphology (Figure 1). Then the ice was chipped away in order to place a bubble trap directly over the identified seep. Short-term measurements consisted of measuring the volume of gas accumulated in submerged bubble traps (corrected for hydrostatic pressure) placed over seeps for periods of ≈ 20 min to up to 6.1 days. The mean and median periods of short-term measurements were 1.6 and 1.0 days per seep, respectively. In most cases, short-term flux measurements were made only once per seep, but several seeps were measured twice over the short term.

[23] Long-term flux measurements consisted of continuously measuring the volume of gas accumulated in submerged traps per unit of time (minutes to weeks); however, these traps were left in place year-round and secured over discrete seeps for periods of up to 359 days in Siberian lakes and up to 700 days in Alaskan lakes. In the Alaskan lakes, semi-automated bubble traps equipped with wet-cup gas meters and event data loggers, similar to those used in other fields of study [Massé *et al.*, 1997], were placed at various depths below the lake surface. We corrected bubble volume data for hydrostatic pressure and lake water temperature. These bubble traps allowed for the volumetric measurement of all gas released from individual seeps over the long period of measurement.

[24] We used unbalanced, type III two-way analysis of variance (ANOVA) to test differences in measured ebullition

among the A, B, C, and Hotspot by lake, region, and measurement period (short versus long) using the statistical software SAS Institute. Degrees of freedom were adjusted to account for any empty cells in an analysis.

2.3. Spatial Statistical Modeling of Ebullition in Lakes

[25] We used data collected from the large survey squares to test three hypotheses for spatial patterns of seeps. Lake bed morphology was tested as a possible explanatory variable for determining seep density. We used point-process models to model seep locations regardless of explanatory variables. Results of point process models were used to create simulations of bubble occurrence in three lakes of the same dimensions as the three intensively studied lakes, Claudi, Goldstream and Ikroavik. Details and results of the spatial statistical modeling are available in Figures S1, S2, and S3 in the supporting information.

2.3.1. Describing Spatial Patterns

[26] We used marked point process models to parameterize the spatial distribution of ebullition seeps in lakes. Marked point processes are widely used in ecological research to analyze data in which the researcher is interested both in the location of an object and also some attribute, or mark [Cressie, 1993]. Here we are interested in the location of seeps and the ebullition class, or mark, of each seep. Three possible formulations exist for a marked point process [Baddeley, 2008]: (1) seeps are equally likely to be located anywhere in a region, and neither seeps of the same class nor seeps of different classes exhibit the interactions of attraction or repulsion (Figure 4a); (2) seeps of different classes are located through independent spatial processes (Figure 4b); or (3) seeps of all classes are located through a single spatial process, and seeps classes are conditionally independent and identically distributed among locations (Figure 4c). Determination of the correct formulation is essential for accurate simulation of point processes [Illian *et al.*, 2008]. To determine which formulation was appropriate for evaluation of ebullition seeps in thermokarst lakes, we tested three null hypotheses [Baddeley, 2008]:

[27] H_{01} : Seeps do not occur in clusters across the lake, and seep class is random (Figure 4a).

[28] H_{02} : The seep classes A, B, C, and Hotspot all follow independent patterns of spatial distribution, and if clustering occurs, the cluster patterns are different among seep classes (Figure 4b).

[29] H_{03} : Ebullition seeps occur in clusters, but within clusters seep class is random (Figure 4c).

[30] To determine which type of point process (H_{01} , H_{02} , or H_{03}) thermokarst-lake ebullition seeps follow, we analyzed spatial seep patterns in the field data sets of Claudi Square, Goldstream Thermokarst Square, Goldstream Center Square, and Ikroavik Square. We determined clustering (an estimate of the spatial dependence between points at a range of scales) by generating K-functions derived from the DGPS locations of seeps [Baddeley, 2008; Van Lieshout and Baddeley, 1999]. All analyses were conducted using the spatstat package [Baddeley and Turner, 2005] in R version 2.12.1 [R Development Core Team, 2010].

2.3.2. Inhomogeneous Poisson Models

[31] Seeps may be considered to occur at higher densities in some zones of a lake due to peculiar properties of that zone. When information about these properties is known,

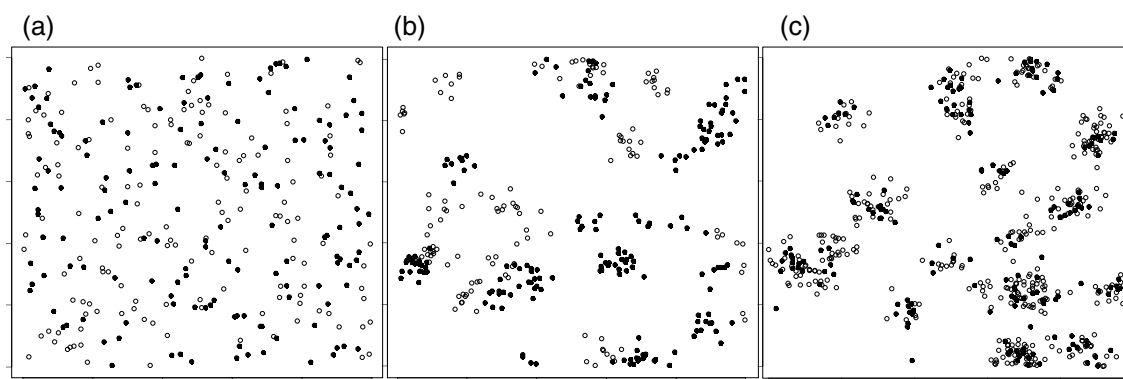


Figure 4. Examples of three hypothetical models for marked point patterns. (a) A point pattern that follows complete spatial randomness (CSR), with marks assigned randomly to points. (b) A point pattern in which two types of points (open and filled) are generated according to separate, independent spatial point processes. (c) A point pattern in which all points are first generated according to a single spatial point process, and then marks are assigned to points through a random mechanism.

inhomogeneous Poisson models may be used to describe seep intensity as a function of one or more of these covariates. To test the hypothesis that seep location is related to lake bed morphology, we fit the inhomogeneous Poisson model with seep intensity modeled as a loglinear function of the lake bed morphology covariate. Models were fit to large square data, where we had identified seep locations and had measured lake bed morphology. We tested lakebed slope and depth as potential morphology covariates. We fit the inhomogeneous Poisson process model using the Berman-Turner algorithm [Berman and Turner, 1992], implemented in the *ppm* procedure from the *spatstat* package in R. The fit of our model was determined using the likelihood ratio test and the Akaike Information Criterion (AIC) [Baddeley, 2008].

2.3.3. Point Process Models

[32] Seep locations were also modeled using a Poisson cluster process model. Poisson cluster processes model the clustering and intensity of points as an intrinsic part of their behavior, without explanatory variables. This model allowed us to describe and simulate seep clustering across whole lakes where precise information on lake bed morphology and other potential covariates is unknown. We fit a Thomas process variant of the generalized Poisson cluster process to the data from each of Claudi Square, the two Goldstream Squares, and Ikroavik Square using the *kppm* procedure from the *spatstat* package in R. The fit of the model was tested using simulation to produce critical envelope values with a 95% significance level for the model [Baddeley, 2008].

2.3.4. Lake Simulation

[33] To test sampling strategies for estimating ebullition in field-based approaches, we created simulated lakes in which the location and efflux from sediments of every ebullition seep was exactly known. To cover a range of thermokarst lake types in Alaska, we simulated seep ebullition across the surfaces of three lakes, producing Lake Claudi_{sim}, Goldstream_{sim} Lake, and Ikroavik_{sim} Lake (Figures 3d–3f) with the same areas and shapes as the lakes studied in field work. Each simulated lake consisted of the known, georeferenced lake surface populated with simulated seeps. We used lake-specific field survey data from squares and transects to determine seep density and survey data from squares to determine spatial parameters for Thomas process

models of seep clustering on each lake. We then used these lake-specific models to simulate seep ebullition across the entire surface of each lake.

[34] We created zones of differing seep intensities on lakes according to generalization of field observations. For Goldstream_{sim} Lake, we defined two zones of differing seep intensity (Figure 3e): a high-intensity thermokarst zone within ≈ 20 m wide belt along the eastern margin and a lower intensity nonthermokarst zone, which was the rest of the lake, using a seep intensity transition line observed in ice-bubble field surveys and early winter aerial photos of Goldstream Lake. In aerial photos, the density of open-hole Hotspots drops off sharply at this boundary, which was approximately twice the distance from the 1949 shoreline to the modern shoreline. If expansion rates in the past were similar to that observed from 1949 to the present, these observations suggest that enhanced methane production and emission occur in response to thermokarst in Goldstream Lake on the order of 100–120 years following the thermokarst conversion of land to lake. Zone-specific field survey data of seep class densities and ratios collected on squares and transects were used to create Thomas process models for simulating seeps separately by zone.

[35] At Lake Claudi, thermokarst expansion is observed in all directions (Figure 3a). For Lake Claudi_{sim}, we defined seep intensity as a continuous variable proportional to the distance from the center of the lake (Figure 3b), consistent with our field observations of seeps [this study] and modeling observations of methanogenesis [Kessler *et al.*, 2012].

[36] At Ikroavik Lake, ebullition seep density was low across the whole lake surface. The lake surface was considered one uniform zone for seep intensity on Ikroavik_{sim} Lake.

[37] After simulating seep locations, we assigned a simulated ebullition flux value (SEF, $\text{mL seep}^{-1} \text{d}^{-1}$) to each simulated seep in each lake. From the PALIMN field data set of flux from 162 lake ebullition seeps, we determined that seep flux follows a lognormal distribution (Figure 5a). To assign SEF values, we created random lognormal distributions for each class of simulated seeps on each simulated lake, pooled these distributions by lake while maintaining a class assignment, and randomly resampled them to produce lake-specific flux distributions for each simulated lake

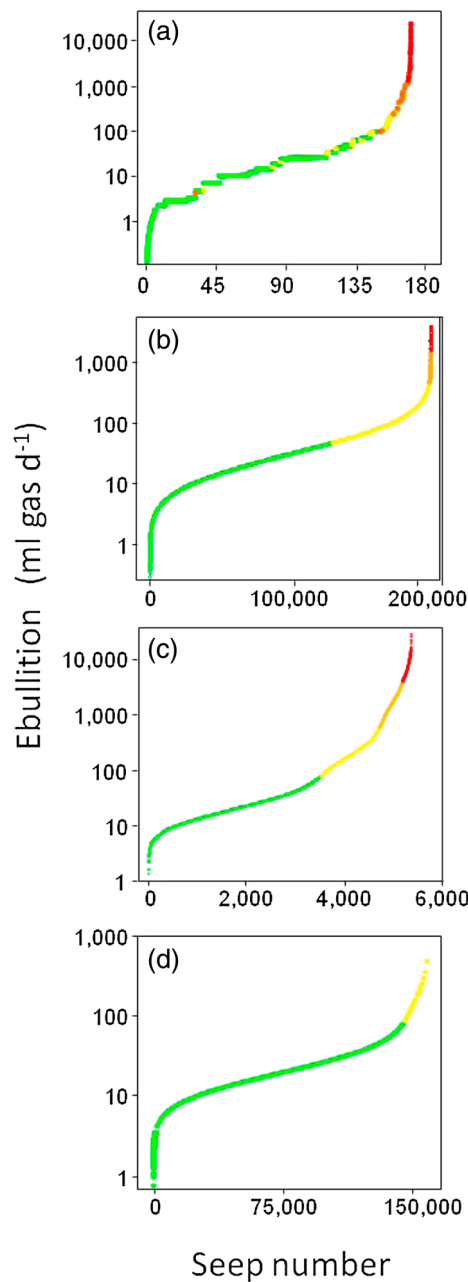


Figure 5. The distribution of ebullition flux of 162 field-measured seeps weighted by the ratios of seep classes observed among 16,364 seeps individually classified and mapped on (a) 75 Alaskan lakes [Walter Anthony *et al.*, 2012] and simulated seeps on (b) Lake Claudisim, (c) Goldstreamsim Lake, and (d) Ikroaviksim Lake. Colored dots represent seep types: A, green; B, yellow; C, orange; and Hotspot, red. Ebullition flux (y axis) followed a lognormal distribution for measured seeps and seeps on simulated lakes, but the number of seeps and ratios of seep classes on each simulated lake differed in accordance with field observations.

(Figure 5b–5d). Simulated fluxes were randomly attached to simulated seep locations.

[38] In addition to having a unique SEF value, each seep was also assigned a PALIMMN flux value, which was one of the four mean fluxes for each seep class (A, B, C, and

Hotspot) in Figure 1. PALIMMN values were assigned based on which seep class the SEF value fell into. This means that on simulated lakes, each seep had two ebullition values in units of $\text{mL seep}^{-1} \text{d}^{-1}$ assigned to it: (1) the PALIMMN class average (Figure 1) and (2) a unique and random SEF value from the lognormal distribution of seep fluxes that occur in lakes. Seep class assignment was retained in order to compare sampling methodologies using mean flux by bubble class (PALIMMN) with methodologies measuring ebullition flux on each seep (SEF).

[39] We also created a separate set of 26 simulated lakes to test for the effects of variation in the percentage of lake surveyed, seep density, and clustering patterns on the accuracy and precision of lake-ice ebullition surveys. Eight of these simulated lakes varied in area with constant seep density and clustering pattern; nine varied in seep density with constant area and clustering pattern; and nine varied in clustering pattern with constant area and seep density (Table 3). The same bubble class ratios and ebullition flux distributions were assigned to seeps in all of these 26 simulated lakes using the methodology described above.

[40] We performed all lake simulations in R [R Development Core Team, 2010], using the *rThomas* procedure from the R spatstat package [Baddeley and Turner, 2005] to create seep locations and using the *rlnorm* procedure in R to create the seep fluxes.

2.4. Sampling Evaluation

[41] Knowledge of the exact location, class, and associated flux of every seep on a simulated lake allowed us to test sampling strategies commonly used in field work. Specifically, we compared the simulated lake fluxes with estimated fluxes from

Table 3. Lake Areas, Seep Densities, and Clustering Parameters for 26 Simulated Lakes

Parameter	Lake	Radius	Area	Seep Density	Clustering Parameters		
		(m)	(m^2)	(seeps m^{-2})	κ	σ	μ
Lake area	1	30	2,815	0.30	0.041	0.758	7.46
	2	50	7,833	0.30	0.041	0.758	7.46
	3	70	15,364	0.30	0.041	0.758	7.46
	4	100	31,375	0.30	0.041	0.758	7.46
	5	200	125,581	0.30	0.041	0.758	7.46
	6	400	502,488	0.30	0.041	0.758	7.46
	7	800	2,010,285	0.30	0.041	0.758	7.46
	8	1600	8,041,807	0.30	0.041	0.758	7.46
Seep density	1	200	125,581	0.02	0.041	0.758	0.50
	2	200	125,581	0.04	0.041	0.758	1.00
	3	200	125,581	0.08	0.041	0.758	2.00
	4	200	125,581	0.12	0.041	0.758	3.00
	5	200	125,581	0.19	0.041	0.758	5.00
	6	200	125,581	0.33	0.041	0.758	8.00
	7	200	125,581	0.52	0.041	0.758	13.00
	8	200	125,581	0.80	0.041	0.758	20.00
Clustering pattern	9	200	125,581	2.06	0.041	0.758	50.00
	1	200	125,581	0.50	0.010	0.758	50.00
	2	200	125,581	0.50	0.020	0.758	25.00
	3	200	125,581	0.50	0.040	0.758	12.50
	4	200	125,581	0.50	0.060	0.758	8.33
	5	200	125,581	0.50	0.080	0.758	6.25
	6	200	125,581	0.50	0.100	0.758	5.00
	7	200	125,581	0.50	0.150	0.758	3.33
	8	200	125,581	0.50	0.200	0.758	2.50
	9	200	125,581	0.50	0.300	0.758	1.67

both transect and floating bubble-trap methods to determine the error associated with these two sampling approaches.

2.4.1. Errors Associated With Estimating Ebullition From Transects on Simulated Lakes

[42] Simulated transects were placed on simulated lakes following the field survey methodology of *Walter Anthony et al.* [2010], whereby transects are evenly distributed and randomly placed along different shorelines and different zones, including the approximate centers of lakes. In the set of 26 simulated lakes, transect origins were randomly placed on the shoreline or in the lake center. Each shoreline transect was oriented toward the center of the lake, and all transects were defined as 50 m in length and 1 m in width.

[43] Evaluation of transects was conducted in ArcGIS 9.3 [ESRI]. Simulated seeps were converted to vector file formats using the maptools package [Lewin-Koh and Bivand, 2011] in R. Seeps were intersected with transects in ArcGIS using the *Intersect* tool. For each iteration, we recorded the SEF flux ($\text{mL m}^{-2} \text{d}^{-1}$), based on the sum of the precise simulated ebullition flux (SEF) of simulated seeps on transects divided by transect area. To empirically determine errors associated with the transect survey method, we calculated the mean ebullition flux for the simulated transects, calculated the transect error as $[E = P_{(\text{predicted flux from transects})} - A_{(\text{actual whole lake mean flux})}]$, and determined the variance of the errors based on 250 iterations of random transect placement for each lake. We estimated the standard deviation of the errors as the square root of the variance. We completed this process for combinations of $n = (1, 2, 3, \dots, 9, \text{ and } 30)$ transects from each of the three transect zones on each simulated lake. We conducted analysis of variance to determine significance of density, clustering, area of lake surveyed and the number of distributed transects surveyed on the relative error of whole-lake ebullition estimates.

2.4.2. Comparison of Floating Bubble-Trap and Transect Methodologies

[44] We evaluated and compared tethered, floating bubble traps with lake-ice transects as methodologies for estimating ebullition. For evaluation of the floating bubble-trap method, we conducted 1000 iterations of random placement of seventeen 0.2 m^2 circular, simulated floating bubble traps on *Claudi_{sim}*, *Goldstream_{sim}*, and *Ikroavik_{sim}* lakes. For each iteration, we determined the total seep ebullition flux captured by the 17 floating bubble traps. For evaluation of the transect method, we conducted 1000 iterations of random placement of three 50 by 1 m transects on each of the simulated lakes. The *Goldstream_{sim}* Lake shoreline was divided into one thermokarst and two nonthermokarst segments, and one transect origin was randomly placed in each segment. Each simulated transect ran perpendicular to the shoreline toward the lake center. For *Ikroavik_{sim}* Lake, where simulated seep density was uniform across the lake, the shoreline was divided into three segments, and one transect origin was randomly placed in each segment. For *Claudi_{sim}* Lake, where simulated seep density was increased from the lake center to the margins, one transect origin was randomly placed within each of two shoreline segments, and one transect origin was placed in the center of the lake, oriented with a random heading. For each iteration, we determined both the SEF and PALIMMN ebullition captured by the three transects as follows:

[45] 1. SEF flux ($\text{mL m}^{-2} \text{d}^{-1}$) was the sum of the precise simulated ebullition flux (SEF) of simulated seeps on transects divided by transect area.

[46] 2. PALIMMN flux ($\text{mL m}^{-2} \text{d}^{-1}$) was the sum of the class-specific mean ebullition rates (PALIMMN, Figure 1) of simulated seeps on transects divided by transect area.

[47] For *Goldstream_{sim}* Lake only, we repeated this process using 5 by 10 m polygons instead of 1 by 50 m transects (Figure 3e) to compare the spatial errors associated with the alternative survey method of 50 m^2 polygons instead of 50 m^2 transects. Results from each iteration were recorded and used to construct sampling distributions for each method [Thompson, 2012]. Simulation of methodologies was conducted using Python scripting in ArcGIS 9.3 [ESRI]. We compared resulting sampling distributions using the two-sample Kolomogorov-Smirnov test in R [R Development Core Team, 2010].

3. Results and Discussion

3.1. Ice Bubble Patterns Represent Statistically Distinct Fluxes

[48] Field-measured ebullition rates for 162 seeps in 24 panarctic lakes identified a priori according to ice-bubble pattern (A, B, C, and Hotspot) varied by seep class (ANOVA, $F = 4.72_{46,115}$, $p < 0.0001$; Figure 1). Interaction effects of seep class by lake or seep class by region were not significant. This indicates that the four different ice-bubble patterns represent statistically distinct mean ebullition fluxes irrespective of the lake or region in which they are measured. The range and magnitude of emissions covered by the four seep classes in Figure 1 are also consistent with observations in other lakes and reservoirs [Del Sontro et al., 2010, 2011; Ramos et al., 2006; Schubert et al., 2012].

[49] While measurement period (long versus short) also did not affect the mean ebullition rates by seep class at $\alpha = 0.05$ level, the range of ebullition values for individual seeps was generally larger among the short-term measured seeps (Figure 6). Short-term ebullition is strongly controlled by hydrostatic pressure dynamics [Martens and Klump, 1980; Mattson and Likens, 1990; Fechner-Levy and Hemond, 1996; Scandella et al., 2011a], so measured ebullition rates vary by orders of magnitude over time scales of hours to days [Glaser et al., 2004; Walter Anthony et al., 2010; Goodrich, 2010; Goodrich et al., 2011]. In contrast, ebullition values reported for the long-term monitored seeps are the average of a large number of shorter-term flux measurements on individual seeps (up to $\approx 61,500$ measurements per seep), so the high and low fluxes from individual seeps measured over the long term converge on a more representative ebullition rate for individual seeps. While a single short-term measurement of a seep is highly unlikely to represent the long-term mean ebullition rate from that seep, the mean ebullition of a large number of seeps measured over the short term at different times and in different locations was not different from the mean ebullition of different seeps measured over the long term (Figure 6). Two caveats should be highlighted. First, short-term measurements of A and B seeps were almost entirely conducted during the early winter ice-cover period following removal of overlaying ice since these low-flux seeps are easiest to locate in lakes when seasonal ice cover reveals their

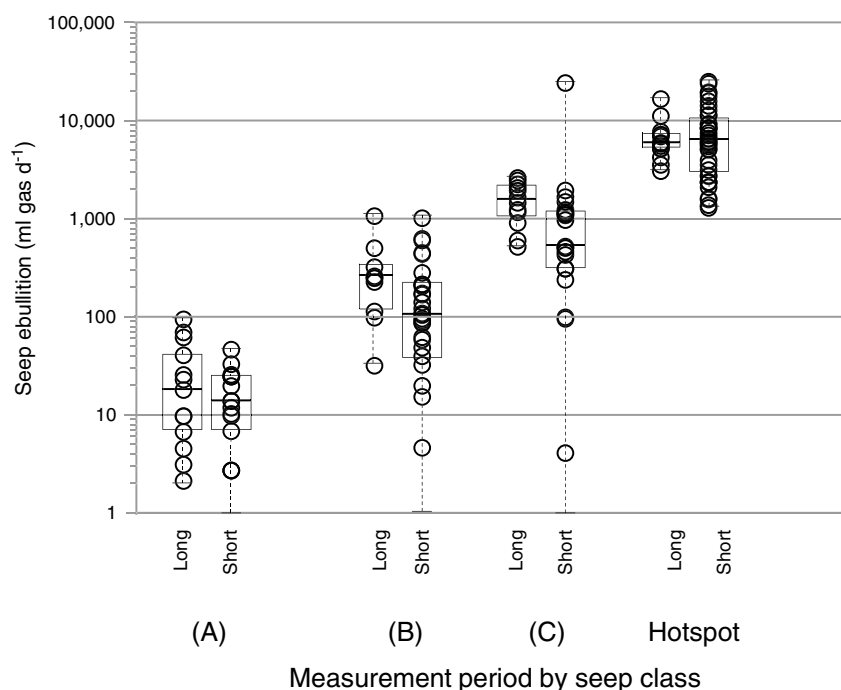


Figure 6. Ebullition fluxes measured on 162 ebullition seeps (open symbols) pre-classified according to ice-bubble patterns as A, B, C, and Hotspot. Fluxes were measured over long (up to 700 days) and short (up to 6 days) periods for different seeps. Box plots show median, maximum, minimum, first quartile, and third quartile values.

locations as bubbles trapped in lake ice. Since low-flux A and B seeps originate from relatively shallow lake sediments [Walter *et al.*, 2008], cold winter temperatures in shallow sediments slow rates of methanogenesis [Zimov *et al.*, 2001; Liikanen *et al.*, 2002; Metje and Frenzel, 2005; Duc *et al.*, 2010], effectively lowering the seasonal efflux from sediments from these seeps. The higher mean ebullition rate reported for long-term monitored A and B seeps takes into account summertime ebullition from these seep classes, which was higher due to warmer temperatures in surface lake sediments in summer [Walter *et al.*, 2006]. Similar responses to seasonal changes in solar bed warming and cooling have been described for other temperature-dependent physical and microbial processes, such as gas solubility and anaerobic oxidation of methane coupled to sulfate reduction in shallow marine sediments [Dale *et al.*, 2008]. Second, while the relative error (standard deviation divided by the mean) was generally under one for A, B, and Hotspot seep classes, regardless of long versus short measurement periods, measurement period had a strong influence on relative error in the C-class ebullition seeps. Long-term relative error of C-class seeps was 0.4, while short-term relative error was 2.4. This large discrepancy is most likely explained by the high-amplitude ebullition dynamic characteristic of C-type seeps. Long-term monitoring of C-type seeps revealed that C-seeps emit bubbles at high rates typical of Hotspots over short periods of time, but they also have long (days to weeks) periods of quiescence. The lack of bubbling over these long periods allows ice to thicken over C-seeps in winter. Temporary interruptions of high-flux ebullition in C-seeps results in large, frequently isolated pockets of gas that stack in between the ice layers. As with all seep classes, long-term monitoring of C-seeps provides representative long-term

ebullition values for these seeps; however, since short-term C-seep ebullition can vary from zero to 10^4 mL gas d^{-1} , the standard deviation of short-term C-seep measurements was high.

[50] We considered it appropriate to use the pooled flux data set consisting of short-term and long-term monitored seeps to (1) populate seeps on simulated lakes with PALIMMN ebullition values, and (2) during field work, apply the PALIMMN ebullition values to unmeasured seeps identified by ice bubble patterns in survey transects for the following reasons: (a) The mean fluxes of A, B, C, and Hotspots seeps were not statistically different between the short-term and long-term data sets; (b) the pooled data set represents a much larger sample size of seeps within each seep class; and (c) seep classes had consistent ebullition rates among a large number of widely distributed arctic lakes.

3.2. Seep Clustering on Thermokarst Lakes

[51] Our tests of H_{01} and H_{02} rejected these two null hypotheses at $\alpha=0.05$ for Claudi Square, the two Goldstream squares and Ikroavik Square (See Figures S1 and S2 in the supporting information). We found (H_1): ebullition seeps exhibit significant clustering for all seep classes; and (H_2): the clustering patterns were the same for classes A, B, C and Hotspot. Our test of H_{03} failed to reject this null hypothesis at $\alpha=0.05$ at all three lakes (See Figure S3 in the supporting information), so we conclude that given the locations of seeps, seep classes were conditionally independent and identically distributed. In other words, seep density is dependent on location, but seep type is not. The clustering of seeps across lake surfaces indicates that within thermokarst lakes there are regular patterns of methane production and transport of bubbles to the sediment-water

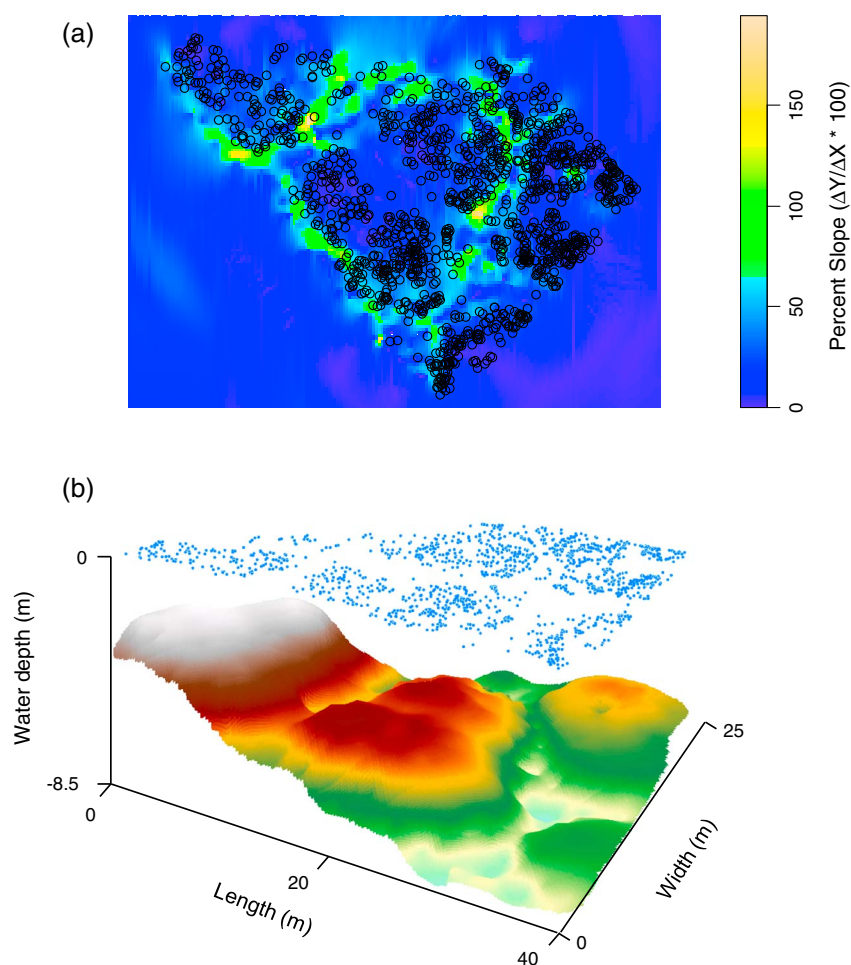


Figure 7. Methane ebullition seep locations from field surveys (a) overlaying slope ratio and (b) bathymetry of Claudi Square lake bed. The location of seeps in lake ice mapped with DGPS is shown as open circles in Figure 7a and as blue dots in Figure 7b. In Figure 7b, lake-bed depth grades from white (−2.5 m) to brown, yellow, green, and turquoise (−8.5 m). Methane seeps were disproportionately associated with the slopes of baydjarakhs, the cone-shaped mounds of thawed permafrost soil distributed among the low-elevation troughs formed by melting of massive ice wedges beneath the thermokarst lake. Baydjarakhs contain the organic matter in thawed permafrost that fuels the majority methane production in yedoma-type thermokarst lakes.

interface and then to the atmosphere. This conclusion allowed us to model seep location irrespective of flux and class; on simulated lakes, it allowed us to first simulate seep locations and then to randomly assign flux to seeps irrespective of location.

3.3. The Spatial Relationship of Ebullition Seeps and Permafrost Thaw

[52] Clustering patterns of seeps in lakes matched the pattern of polygonal ground observed in the permafrost-dominated regions. In field work in the tundra and boreal yedoma permafrost study regions, we observed that dense clusters of methane seeps were distributed ≈ 10 m apart across lake ice surfaces. The ice wedge polygons exhibited identical ≈ 10 to 12 m spacing. As sedimentation and soil formation progress in the polygon centers over millennia, surrounding ice wedges syngenetically grow both vertically with the sedimentation and in width by repeated frost cracking and refilling with vein ice. If a disturbance to the ground

surface occurs that causes permafrost to thaw, ice wedges melt and the soils of the polygon center remain as slowly eroding conical thermokarst mounds known as baydjarakhs [Czudek and Demek, 1970; Kanevskiy *et al.*, 2011]. Baydjarakhs form a regularly rough topography, similar in morphology to the bottom of an egg carton, but with the spacing and distribution of the former ice wedge polygons. We observed 10 to 12 m spacing of subaerial baydjarakhs near lake margins. Bathymetric maps created by sonar depth measurements beneath Claudi Square (Figure 7), Goldstream Thermokarst Square (not shown), and other linear survey transects in these lakes revealed that this ≈ 10 m spacing of baydjarakhs continued beneath the lake, on the yedoma-lake bottoms.

[53] Were methane bubbles seeping out of the tops of these baydjarakh mounds, or from the troughs in between mounds? In most lakes, sediments and organic detritus are focused toward the deep centers of basins [Lehman, 1975]. In thermokarst lakes, sediment focusing occurs at multiple

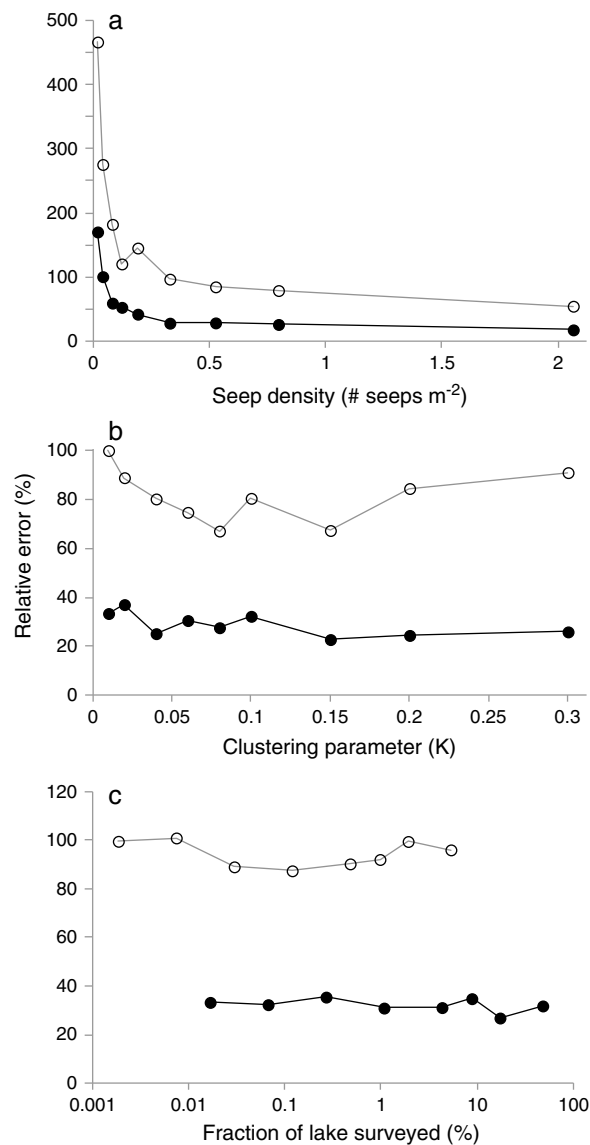


Figure 8. The relative error (%) of the mean whole-lake ebullition estimate based on one (open symbols) and nine (closed symbols) transects per sector randomly distributed on 26 simulated lakes that varied according to (a) seep density, (b) seep clustering, and lake size, which translates to fraction of the lake surveyed based on 50 m² (1 by 50 m) transects. Simulated ebullition fluxes (SEF) were unique ebullition values assigned to each seep randomly on the 26 simulated lakes following the distribution of fluxes measured on 162 seeps weighted by the ratios of these classes as they occurred among 16,364 seeps individually classified and mapped on 75 Alaskan lakes (Figure 5a) [Walter Anthony *et al.*, 2012].

spatial scales, both toward the deep central basin of the lake [Hopkins and Kidd, 1988; Murton, 1996], but also toward the topographical low points where ice wedges melted forming troughs in between baydjarakhs [personal observation from lake coring and paleolake exposure sampling, unpublished data; Farquharson, 2012]. We used spatial statistics and radiocarbon dating to determine whether seeps

were associated with baydjarakhs (consisting of organic-rich thawed permafrost soil) or troughs (topographically low points for focusing organic-rich surface sediments), both potential sources of organic matter to fuel methanogenesis.

[54] The inhomogeneous Poisson model indicated that lake bed morphology was highly associated with seep location. Seep intensity in the field data sets was significantly related to baydjarakh slope (χ^2 test, $p < 0.0001$). Adding depth to the model did not improve the model fit. Seep intensity was highest near flatter tops of baydjarakhs and lowest on the steep baydjarakh sideslopes (Figure 7). Baydjarakh tops are very broad (6–8 m), while the trough bottoms are quite narrow (0.5 m), so that >90% of ebullition occurs from the tops of baydjarakhs. Radiocarbon ages of methane collected from bubbles trapped in ice in the dense bubble clusters above two separate baydjarakhs closest to the thermokarst margin in Claudi Square were 40,400 and 41,300 years. The association of seep clusters with baydjarakhs and the radiocarbon ages of methane that matched those of the Pleistocene-aged organic matter in yedoma permafrost are both pieces of evidence that support the hypothesis that permafrost thaw fuels methane production in thermokarst lakes, particularly lakes formed in icy yedoma permafrost.

[55] Sonar surveys of Goldstream Lake and Lake Claudi revealed consistent, lake-wide baydjarakh patterns in the lake bed morphology. The significant relationship between lake bed morphology and seep density allowed us to apply the seep clustering observed in polygon samples to the entire lake surfaces.

3.4. Modeling Seeps as a Point Process to Quantify Uncertainty in Lake Emission Estimates

[56] Because we rejected H_{01} and H_{02} , but failed to reject H_{03} , we modeled ebullition seep location as a single point process and randomly assigned seep classes based on field observations of class flux distributions (Figure 5) once seep locations were determined. The Thomas process model also provided excellent fit of the seep location data over seep separation distances of 0 to 8 m, indicating that the clustering of seep locations could be modeled without explanatory data from lake bed covariates. The resulting distribution of seeps in the simulated lakes, Lake Claudi_{sim}, Goldstream_{sim} Lake, and Ikroavik_{sim} Lake (Figures 3d–3f), were consistent with our field observations of ebullition in the actual lakes; however, unlike in the actual lakes, the location and flux of all ebullition seeps on simulated lakes was known exactly.

[57] Knowing the precise simulated ebullition flux (SEF) value for each individual seep on the 26 simulated lakes (Table 3), we could distribute different numbers of transects and polygons across simulated lakes to determine the error associated with estimating whole-lake methane ebullition based on limited spatial sampling. Increasing the number of transects from one transect per zone to nine transects per zone reduced the relative error associated with estimating whole lake ebullition at the 95% confidence level within all lakes (Figure 8). Among lakes, contrary to our initial expectation, differences in the uncertainty of ebullition estimates using the survey transect method were not related to the fraction of the lake surveyed or to the seep clustering parameter (Figure 8 and Table 4). Rather, we found that uncertainty in upscaling transect data to whole-lake ebullition flux values

Table 4. Results of ANOVA for Prediction of Relative Error in Whole-Lake Ebullition Estimation

Variable and Source of Variation	d.f.	MS	F	P
Seep density	1	11.3	9594.27	<0.001
Seep clustering	1	0.0	2.39	0.124
Percentage of lake surveyed	1	0.0	0.21	0.645
Number of transects	1	5.9	5048.06	<0.001
Seep density $\times$ number of transects	1	1.8	1568.10	<0.001

was inversely related to seep density (Figure 8a, $R^2=0.99$). Equation (1) defines the relative error RE at the 95% confidence level associated with the mean lake ebullition estimate as a function of seep density D and the number of distributed survey transects T :

$$RE = 0.6866D^{-0.481}T^{-0.51} \quad (1)$$

[58] On lakes with low seep density, the probability was low that an individual transect would intersect a representative number of seeps. In contrast, on lakes with high seep density, the probability was much higher that an individual transect would intersect a representative number of seeps. Increasing the number of transects surveyed reduced uncertainty in whole-lake ebullition estimates by increasing the likelihood that transects represent the spatial heterogeneity in the density of ebullition seeps across lakes (Table 5). This concept is highlighted by the results shown in Figure 8c: multiple dispersed transects result in lower errors than a single transect, even when the fraction of the lake surveyed is the same. These results are consistent with the findings of *Wik et al.* [2011a], who also found that larger numbers of widely distributed ebullition survey transects reduce uncertainty in seep ebullition flux estimates on two lakes in northern Sweden.

3.5. Comparison of Ice-Bubble Survey Transects Versus Polygons

[59] In the comparison of an alternative survey method of equal area, using 5 by 10 m polygons as opposed to 1 by 50 m linear transects, we found that surveying polygons on Goldstream_{sim} nearly doubled the relative error associated with the mean flux estimate of lake ebullition. The high variance associated with polygon surveys resulted from similarity in the scale among polygons, seep clustering, and baydjarakhs. While a 50 m long transect was likely to intersect several baydjarakhs and associated seep clusters, polygons were apt to fall on or between baydjarakhs and seep clusters, giving rise to large variances. These results imply that for field surveys, the distributed 50 m transect method is preferred to more nearly isometric polygons of equal area. Long, continuous transects are more likely to cover decameter-scale heterogeneity in seep distribution within and among zones in the lake. Transects are also logistically easier to survey in the field because walking along a 1 m wide strip is more efficient and reduces the likelihood of missing or double counting seeps, a challenge that is present in surveying polygons.

3.6. Comparison of Ice-Bubble Survey Transects and Bubble-Trap Methods

[60] In the comparison of alternative survey methods for estimating seep ebullition, using 0.2 m² floating bubble traps as opposed to 1 by 50 m linear transects, we found that 0.2 m² floating bubble traps resulted in substantial underestimation of the mean flux. The magnitude of this underestimation ranged from 1.5-fold to >100-fold, depending on seep density. For Goldstream_{sim} Lake, where seep density was near the middle of the observed range, floating bubble traps produced a median estimate of whole-lake ebullition that was 11.5 times less than the actual mean whole-lake ebullition and 10.9 times less than the estimate produced by the linear transects. The classic bubble-trap method also underestimated mean lake flux in 63–90% of occasions (Figure 9), also depending on seep density. The sampling distribution from the floating bubble-trap method exhibited extreme deviation from normality, indicating substantial median-bias and associated difficulties with statistical inference from this method if it were used in the field. In contrast, the linear transect method underestimated mean lake ebullition by a much smaller amount (5–34%) and produced sampling distributions that approximated normality. We attribute the poor performance of the floating bubble-trap method to well-described problems encountered in sampling of rare events [Thompson, 2012]. Given that the mean seep density on the simulated lakes was 0.5 seeps m⁻² and that each floating bubble trap covered 0.2 m², there existed a 0.9 probability that individual bubble traps would capture no ebullition seeps. As a result, the sampling distribution of the floating bubble traps was highly skewed toward underestimation of ebullition flux. This magnitude of underestimation for floating bubble traps is similar to that observed by *Wik et al.* [2011a, 2011b] and *Walter et al.* [2006] in comparisons of these two methods in the field.

Table 5. The Relative Error (RE, Percent of the Mean) for the 95% Confidence Level of Mean Lake Ebullition Based on Equation (1)^a

Seep Density (# seeps m ⁻²)	Number of Transects						
	3	6	9	15	21	27	90
0.01	343.8	234.8	187.9	141.9	117.9	102.7	53.0
0.03	202.7	138.4	110.8	83.6	69.5	60.5	31.2
0.05	158.5	108.3	86.6	65.4	54.4	47.3	24.4
0.07	134.8	92.1	73.7	55.6	46.2	40.3	20.8
0.09	119.5	81.6	65.3	49.3	41.0	35.7	18.4
0.2	81.4	55.6	44.5	33.6	27.9	24.3	12.5
0.4	58.3	39.8	31.9	24.1	20.0	17.4	9.0
0.6	48.0	32.8	26.2	19.8	16.5	14.3	7.4
0.8	41.8	28.5	22.8	17.2	14.3	12.5	6.4
1	37.5	25.6	20.5	15.5	12.9	11.2	5.8
1.2	34.4	23.5	18.8	14.2	11.8	10.3	5.3
1.4	31.9	21.8	17.4	13.2	10.9	9.5	4.9
1.6	29.9	20.4	16.4	12.4	10.3	8.9	4.6
1.8	28.3	19.3	15.5	11.7	9.7	8.4	4.4
2	26.9	18.4	14.7	11.1	9.2	8.0	4.1

^aWhere RE is a function of seep density and number of distributed transects per lake surveyed. The transect dimensions used to statistically derive these relationships were 1 m $\times$ 50 m (50 m²), parameters that follow the PALIMMN protocol [Walter Anthony et al., 2010] and capture the within-zone variability in ebullition controlled by lake bed morphology (10 m scale; section 3.3).

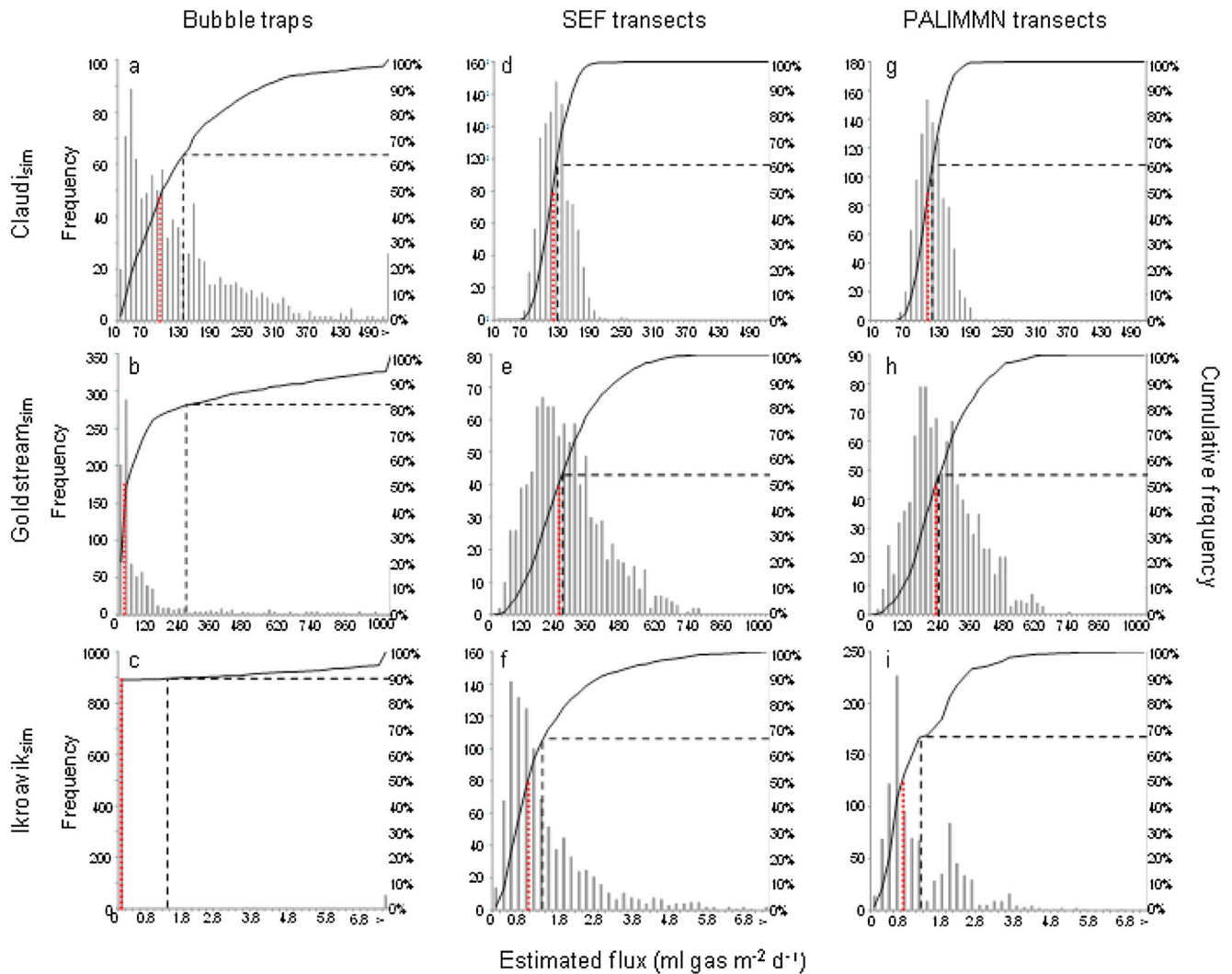


Figure 9. Sampling distributions for estimates of lake ebullition obtained through 1,000 iterations of three sampling methodologies using (a–c) 17 tethered 0.2 m² bubble traps and seep-specific flux [Wik *et al.*, 2011a, 2011b]; (d–f) three distributed 50 m transects and seep-specific flux, SEF; and (g–i) three distributed 50 m transects and PALIMMN-estimated flux by class (Figure 1), applied to simulated ebullition seeps on three lakes of different seep density (Claudisim, 1.23 seeps m⁻²; Goldstreamsim, 0.53 seeps m⁻²; and Ikroaviksim, 0.03 seeps m⁻²). Gray bars are a histogram of the sampling distribution of estimated flux. Solid black line indicates the cumulative sampling curve. Vertical dashed black line indicates the actual mean flux for the simulated lake; horizontal dashed black line indicates the intersection of actual mean flux with the sampling distribution. Vertical dashed red line indicates the median estimated flux for the sampling distribution. The bubble-trap method produced median flux estimates that were 1.5, 11.5, and >100 times lower than actual mean lake fluxes on Claudisim, Goldstreamsim, and Ikroaviksim, respectively, and underestimated actual mean flux in 64, 80, and 90% of occasions for these lakes. The sampling distributions of the bubble trap method exhibited extreme departure from normality and high median-bias toward underestimation of flux. Ice-bubble survey transect methods based on SEF and PALIMMN seep values produced median flux estimates that were 1.09, 1.06, and 1.6 times lower than actual flux for lakes Claudisim, Goldstreamsim, and Ikroaviksim, respectively, and underestimated actual flux in 60, 55, and 66% of occasions for these lakes. The sampling distributions of ice-bubble survey transect methods were much closer to normal distributions than those of the bubble trap method. All methods produced more narrowly distributed sampling estimates with increasing seep density. The ice-bubble survey transect methods produced results that were substantially more accurate than those of the bubble-trap method. Differences between results based on SEF and PALIMMN transect methods were not significant.

3.7. Comparison of Ice-Bubble Transect Estimates Based on Class-Mean Fluxes With Exact Fluxes

[61] There were no significant differences between sampling distributions based on applying simulated seep flux values (SEF) along transects versus applying the mean PALIMMN flux for seep classes on lakes *Claudi_{sim}* and *Goldstream_{sim}* (Figure 9). On the very low seep-density lake, *Ikroavik_{sim}*, three transects intersected very few seeps (mode=4), resulting in a bimodal sampling distribution when mean PALIMMN flux values were applied. There was no indication from values of the mean, median, or variance that this resulted in decreased accuracy. When sampling intensity on *Ikroavik_{sim}* was increased to nine transects, we observed no significant differences between SEF and PALIMMN sampling distributions (data not shown). Because there is considerable variability in seep flux within class (Figures 5 and 6), applying the individual, seep-specific SEF values induced as much error as it resolved. This is a highly important result given that determining the true flux of individual seeps along transects is logistically impractical in extensive field work. The implication of this finding is that PALIMMN flux values can be applied to seeps by class during field surveys without affecting the accuracy of the whole-lake mean seep ebullition estimate.

3.8. Quantifying Uncertainty in Field-Based Lake Ebullition Estimates

[62] In the past, upscaling spatially limited transect-level field data to whole-lake ebullition estimates was inhibited by a lack of knowledge of the relative errors associated with mean flux estimates from transects. In this study, spatial analysis of randomly placed transects across a variety of simulated lakes, for which whole-lake ebullition was precisely known, revealed the critical relationship between the number of distributed survey transects and the density of seeps on a lake to constrain the relative errors associated with mean ebullition estimates from transects (section 3.4 and Figure 8a). Based on this relationship (equation (1)), the relative error values predicted for different combinations of seep densities and distributed 50 m² survey transects are provided in Table 5.

[63] If there is no prior knowledge about the areal distribution of processes that control ebullition in a lake, then survey transects should be randomly distributed in lakes, and ebullition estimated as the cumulative seepage from all surveyed areas divided by the total area surveyed. Following this approach at *Ikroavik Lake*, the total amount of ebullition observed among the 32 survey transects was 5227 mL d⁻¹. Dividing this flux by the total area surveyed (1873 m²) yielded a mean flux of 2.8 mL m⁻² d⁻¹. To determine the uncertainty of this mean flux estimate, we used the mean density of seeps on *Ikroavik* (0.07 seeps m⁻²) and the number of transects surveyed in the lake (32 transects) in equation (1) to calculate the relative error (37%). Propagating this spatial sampling error together with the flux-based relative error (26%) associated with seep-class mean ebullition rates from Figure 1, yielded a mean whole-lake seep ebullition estimate for *Ikroavik Lake*, 2.8 ± 1.3 mL m⁻² d⁻¹ at the 95% confidence level. Following the same approach for *Lake Claudi*, where the mean seep density

was 1.74 seeps m⁻² along 7 transects, the relative error on the mean estimate of whole-lake ebullition was 25%, yielding a whole-lake mean ebullition estimate of 152 ± 38 mL m⁻² d⁻¹ at the 95% confidence level.

[64] In *Goldstream Lake*, since 11 transects were representatively distributed across two, well-constrained zones (thermokarst and nonthermokarst), we determined the relative error (24%) based on the propagation of errors associated with the mean seep density, 0.78 seeps m⁻² using equation (1) and the flux-based errors (Figure 1). This approach, which excluded the two large surveyed polygons in the thermokarst and nonthermokarst zones, yielded a whole-lake ebullition estimate of 254 ± 62 mL m⁻² d⁻¹. Separately, we used a weighted mean for only those transects that were fully in the thermokarst or nonthermokarst zones together with the large polygons that were fully in these zones, considering that these zones occupied 18% and 82% of the lake area, respectively. This weighted approach yielded a mean whole-lake ebullition estimate, 253 ± 82 mL m⁻² d⁻¹, similar to that of the transect-only method.

[65] Using equation (1) and the flux-based errors of each seep class to estimate the relative error in the mean estimate of whole winter flux for two lakes in northern Sweden studied by *Wik et al.* [2011a] (mean flux 87 mg CH₄ m⁻² d⁻¹, mean seep density 1.63 seeps m⁻², surveyed area equivalent to 5.5 l m × 50 m transects), we calculate an estimate of 87 ± 25 mg CH₄ m⁻² d⁻¹ at the 95% confidence level. Propagating the variances presented by *Wik et al.* [2011a] to calculate a relative error in the estimate of the mean [Ku, 1966], we arrive at 87 ± 33 mg CH₄ m⁻² d⁻¹ at the 95% confidence level. The close agreement between our simulation-based error estimates and the estimates derived from *Wik et al.* [2011a] provides corroborating evidence that ice-bubble transect surveys yield well-constrained estimates of mean lake ebullition on the Swedish study lakes.

3.9. Mass-Based Methane Ebullition From Seeps

[66] To convert volumetric ebullition to a mass-based estimate of methane emissions to the atmosphere, the methane concentration in ebullition bubbles at the surface of the lake must be known. On *Goldstream Lake*, methane concentrations (percent by volume) based on 246 measurements of 42 ebullition seeps (type A = 82 ± 3%, n = 6; type B = 83 ± 7%, n = 3; type C = 85 ± 1%, n = 14; type Hotspot = 89 ± 1%, n = 19; reported as mean ± standard error, n number of seeps) were higher than those reported as the mean from a much larger set of arctic and subarctic PALIMMN lakes (A, 73%; B, 75%; C, 76%; and Hotspot, 78%) [Walter Anthony et al., 2010]. Applying these site-specific methane concentrations measured in bubbles at the surface of *Goldstream Lake* to the volumetric ebullition survey data yielded a mass-based flux estimate for the whole lake, 157 ± 51 mg CH₄ m⁻² d⁻¹ at the 95% confidence level. At *Lake Claudi*, where the bubble gas methane concentration [B = 72 ± 4%, n = 4; C = 73 ± 4%, n = 3] was similar to that of the PALIMMN values [Walter Anthony et al., 2010], and in *Ikroavik Lake*, where we did not measure bubble methane concentration, we applied PALIMMN mean concentrations. This resulted in mass-based flux estimates of 81 ± 20 mg CH₄ m⁻² d⁻¹ and 1.5 ± 0.7 mg CH₄ m⁻² d⁻¹, respectively. It should be noted that these estimates represent the magnitude of mean annual methane ebullition that

reaches the lake surface or bottom of the lake-ice depending on season; so they do not represent actual emissions to the atmosphere. The latter requires consideration of seasonal variability in seep ebullition, ice phenology, and other physical and biological processes that influence methane cycling in lakes.

[67] One contributor to the higher methane concentration values in Goldstream Lake is the relatively shallow water column at this site (2.9 m), which limits the magnitude of degassing from ascending bubbles [McGinnis *et al.*, 2006]. Differences in bubble size could also influence the magnitude of methane dissolution during ascent [McGinnis *et al.*, 2006]; however, bubble sizes did not appear to be different when we made flux measurements or when individual bubbles were trapped by ice. We did not collect quantitative bubble size data at all lakes. Since methane solubility is temperature dependent, sediment and water column temperature will also influence bubble methane content seasonally [Del Sontro *et al.*, 2010]. Similarly, bubbles resting under winter ice prior to entrapment when ice thickens around them are subject to methane dissolution followed by potential microbial oxidation in the water column [Walter *et al.*, 2008; Boereboom *et al.*, 2012]. This process effectively lowers methane ebullition emissions when ice melts in spring since not all methane released from sediments as bubbles reaches the atmosphere. Other biogeochemical cycling process must also influence seep-bubble methane concentrations, because bubbles of similar size released from lakes with similar depths and temperature regimes had highly variable methane concentrations in the PALIMMN data set [Walter Anthony *et al.*, 2010]. To improve accuracy in mass-based ebullition flux estimates, ebullition seep gases should be sampled seasonally, analyzed for methane concentration, and accounted for in whole-lake flux calculations.

3.10. Implications for Estimating Regional Lake Methane Ebullition Using the Example of Lakes Near Toolik Field Station, Alaska

[68] If lakes in a region have commonalities in the physical and biogeochemical processes that control methane production and ebullition emission such as substrate availability, water depth, permafrost characteristics, periglacial history, and lake developmental cycles, then data from multiple lakes can be pooled as if they were multiple transects belonging to a single lake to determine a regional lake flux estimate. For example, the mean and standard error of seep density on lakes near Toolik Field Station was 0.66 ± 0.11 seeps m^{-2} . The total ebullition flux from all 64 transects was 96.8 L d^{-1} surveyed over $2,996 \text{ m}^2$, yielding a mean ebullition flux value of $32 \text{ mL m}^{-2} \text{ d}^{-1}$. Pooling these transect data as if they were collected from a single large regional lake, the relationship between seep density and the number of distributed transects surveyed (equation (1)) suggests that the spatial relative error associated with this ebullition estimate is 8.5%. Propagating this error with the flux-based errors of each of the seep classes in Figure 1 (24%), we conclude that at the regional scale, there is 95% confidence that the mean seep ebullition in lakes in the northern foothills of the Brooks Range near Toolik Field Station is $32 \pm 8.4 \text{ mL m}^{-2} \text{ d}^{-1}$. This exercise demonstrates that pooling results of a low number of transects per lake on a larger number of widely distributed lakes reduces the error associated with

the mean, allowing for better-constrained regional lake ebullition flux estimates.

4. Conclusions

[69] Constraining the magnitude of methane ebullition in thermokarst lakes is important for understanding the role of permafrost thaw in the arctic carbon cycle and feedbacks to global climate change. Until now, the uncertainty in estimates of mean lake ebullition associated with spatial heterogeneity of ebullition seeps was unknown and hindered the upscaling of transect-level seep surveys to whole-lake ebullition estimates. We used short- and long-term ebullition flux measurements and point process modeling to quantify the errors associated with field survey methods and to reveal spatial patterns of ebullition in a variety of thermokarst lakes in Alaska. Our findings indicate that the classic randomized bubble-trap method for estimating mean lake ebullition is highly median-biased toward underestimation of flux. The ice-bubble survey transect method yielded more accurate estimates of methane ebullition in lakes than the classic bubble-trap method and were only slightly median-biased toward underestimation of flux. These results explain previously observed discrepancies between ebullition estimates based on randomly placed bubble traps and estimates based on ice-bubble transects [Wik *et al.*, 2011a, 2011b; Walter *et al.*, 2006]. Class-mean ebullition fluxes measured in the field by the Pan-Arctic Lake Ice Methane Monitoring Network (PALIMMN) can be assigned to seeps identified as A, B, C and Hotspot types according to their ice-bubble patterns in other northern lakes without compromising the accuracy or precision of ice-bubble survey transect estimates of whole-lake mean ebullition emissions.

[70] Field surveys revealed that ebullition was higher and better constrained in the yedoma-type thermokarst lakes that had high seep densities. In these lakes, methane seepage was greatest adjacent to thermokarst margins. Seeps were clustered around conical thermokarst mounds (baydjarakhs) protruding from the lake bottom with a spacing of $\approx 10 \text{ m}$. Soil mounds are the ice wedge polygon centers and thus part of a regular micro scale landscape pattern across permafrost regions, which makes their distribution (including on lake bottoms) also predictable. Radiocarbon ages of ebullition methane, 40.4 kyr and 41.3 kyr, collected as bubbles emitted from the mounds, revealed that Pleistocene-aged organic matter contained in baydjarakhs fuels methanogenesis in the lake bottom.

[71] Statistical results showed that increasing the number of survey transects distributed widely across lake surfaces significantly reduces the uncertainty in whole-lake ebullition estimates. Neither the fraction of the lake surveyed nor seep clustering had a significant effect on the accuracy of whole-lake mean ebullition estimates. However, precision in estimating ebullition seep fluxes at the whole-lake scale using the ice-bubble transect method was inversely related to the density of seeps on lakes. A calculated uncertainty can be applied to estimating ebullition from lakes by propagating errors associated with seep-class mean fluxes, the number of distributed transects surveyed and seep densities. Regional-scale lake ebullition estimates can also be better constrained by pooling results of a low number of transects per lake on a larger number of widely distributed lakes, as

demonstrated by the example of Toolik Field Station area lakes. Such large-scale, bottom-up estimates for regional ebullition emissions could be useful for earth-system modeling [Gao *et al.*, 2013] and for calibrating efforts to detect and quantify methane emissions from lakes using airborne (P. Regmi *et al.*, manuscript in preparation, 2013) and satellite remote sensing [Engram *et al.*, 2012].

[72] **Acknowledgments.** We thank the following persons for assistance in field data collection: Lawrence Plug, Guido Grosse, Prajna Regmi, Ben Jones, Melanie Engram, Louise Farquharson, Chris Arp, Laura Oxtoby, Laura Brosius, Dragos Vas, Marie Geai, Laurel McFadden, Casey Pape, Odin Miller, Josefine Lenz, and Amy Strohm. We thank Ben Jones and Melanie Engram for providing the modern and 1950s lake shoreline data sets. Joy Clein conducted quality control on field data sets. Jeffrey Chanton conducted laboratory isotope analyses. Melanie Engram wrote the Python script for simulating transect and bubble trap placement. We are grateful for logistical support in Interior Alaska from the Water and Environmental Research Center in the Institute of Northern Engineering at UAF, on the northern Seward Peninsula from CH2M Hill Polar Services Fairbanks Office and Bering Air, and on the North Slope from Doug Whiteman, the Barrow Arctic Science Consortium, and the Ukepeagvik Inupiat Corporation. We thank the National Park Service Fairbanks Office for permits to do fieldwork in the Bering Land Bridge National Preserve. This work was supported by NSF OPP grant 0732735, NASA Carbon Cycle Sciences grant NNX11AH20G, the NASA Astrobiology Institute's Icy Worlds node, and the IARC Collaboration Studies with JAMSTEC and JAXA. Ronald Barry, Guido Grosse, and Melanie Engram provided constructive comments on the manuscript.

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